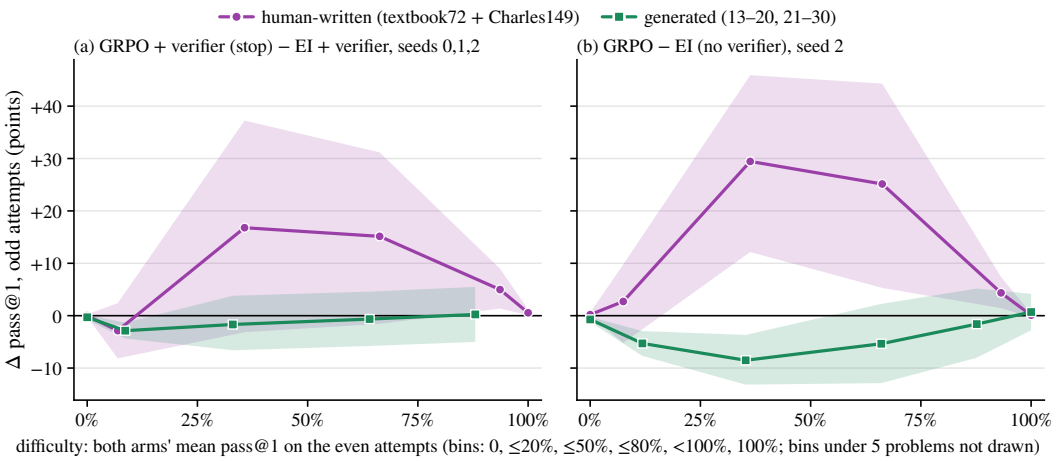
human-written (textbook72 + Charles149)
generated (13–20, 21–30)
(a) GRPO + verifier (stop) − EI + verifier, seeds 0,1,2
(b) GRPO − EI (no verifier), seed 2
Δ pass@1, odd attempts (points)
+40
+30
+20
+10
0
−10
0%
25%
50%
75%
100%
difficulty: both arms' mean pass@1 on the even attempts (bins: 0, ≤20%, ≤50%, ≤80%, <100%, 100%; bins under 5 problems not drawn)